

Lesson Plan: Introduction to Paper Circuits

Overview

Title: Introduction to Paper Circuits **Grade Level:** Elementary to Middle School (Grades 3-8, adaptable) **Subject Areas:** STEAM (Science, Technology, Engineering, Art, Mathematics) **Duration:** 45-60 minutes **Description:** This lesson introduces students to basic electrical circuits using everyday materials like paper, copper tape, LEDs, and coin cell batteries. Students will create simple paper-based circuits that light up, combining hands-on engineering with creative art. The activity emphasizes concepts like conductivity, polarity, and circuit completion while encouraging artistic expression, such as designing light-up cards or drawings.

Learning Objectives

By the end of this lesson, students will be able to:

- Explain the basic components of a simple electrical circuit (power source, conductor, load).
- Identify conductive and non-conductive materials.
- Construct a functional paper circuit that lights an LED.
- Troubleshoot common circuit issues, such as incomplete connections or reversed polarity.
- Apply creativity to integrate circuits into artistic projects, fostering STEAM integration.

Standards Alignment

- **NGSS (Next Generation Science Standards):** 4-PS3-2 (Energy transfer in electrical circuits), MS-PS2-3 (Electric currents and magnetic fields).
- **ISTE Standards:** Innovative Designer (Students use a variety of technologies to create new, useful, or imaginative solutions).
- **Common Core:** Speaking and Listening (Present findings clearly), Mathematics (Measurement and geometry in design).

Materials Needed

(Per student or small group of 2-3)

- Cardstock or heavy paper (for the base, e.g., 8.5x11 inches).
- Copper foil tape (conductive, with adhesive backing; about 1-2 feet per student).
- Coin cell batteries (CR2032, 3V; 1-2 per student).
- LEDs (small, 5mm; various colors; 1-3 per student). Note: LEDs have polarity—longer leg is positive (+), shorter is negative (-).
- Clear tape or glue sticks (for securing components).
- Scissors.
- Markers, crayons, or colored pencils (for decorating).
- Optional templates: Printable paper circuit designs (e.g., simple switch, light-up card, or shapes like a heart or star).
- Optional advanced materials: Binder clips (as switches), aluminum foil (for testing conductivity), or additional LEDs for parallel circuits.

Teacher Prep: Print templates if using (available online or create simple ones with battery and LED placements). Test materials to ensure batteries are fresh and LEDs work. Set up a demonstration circuit.

Safety Considerations

- LEDs and batteries are low-voltage and safe, but remind students not to put batteries in their mouths.
- Scissors should be used with supervision for younger students.
- No open flames or high-heat tools are involved.

Procedure

1. Introduction (10 minutes)

- **Engage:** Begin with a hook—show a pre-made light-up paper card or drawing and ask, "How do you think this lights up without wires or plugs?" Discuss everyday examples of circuits (e.g., flashlights, toys).
- **Teach Key Concepts:**
 - Explain electricity basics: Electricity flows in a loop (circuit) from a power source (battery) through conductors (copper tape) to a load (LED) and back.
 - Introduce terms: Positive (+) and negative (-) terminals; conductivity (materials like metal allow electricity to flow; paper does not unless taped with copper).
 - Demo: Show a simple circuit on paper—place copper tape from battery's + side to LED's + leg, then from LED's - leg back to battery's - side. Press to complete and light the LED.
- **Discussion Questions:** What happens if the circuit is broken? Why does the LED have two different leg lengths?

2. Guided Activity: Building a Basic Paper Circuit (15-20 minutes)

- **Step-by-Step Instructions:**
 1. **Prepare the Base:** Provide each student with paper or a template. If using a template, it should have pre-drawn lines for copper tape paths, battery placement, and LED holes.
 2. **Plan the Design:** Have students sketch a simple drawing (e.g., a flower, robot, or greeting card) and decide where the LED will go (e.g., as an eye or light bulb).
 3. **Apply Copper Tape:**
 - Peel and stick copper tape along the planned paths. Ensure paths don't cross unless intended (crossings need insulation like regular tape).
 - Leave gaps for the LED and battery.
 4. **Add the LED:**
 - Poke small holes in the paper where the LED legs will go.
 - Bend LED legs at 90 degrees and insert through the holes so the bulb is on the front.
 - Tape the longer (+) leg to the copper path from the battery's + side; shorter (-) leg to the - path. Ensure secure connections—press firmly.
 5. **Add the Battery:**

- Place the coin cell battery in its designated spot (e.g., a folded paper pocket).
 - Use copper tape or clips to connect: + side (smooth) to + path, - side (textured) to - path.
6. **Test the Circuit:** Press the battery down if using a simple switch (e.g., a flap of paper). The LED should light up!
- **Troubleshooting Tips (Share with Students):**
 1. LED doesn't light? Check polarity—flip the LED if needed.
 2. No connection? Ensure copper tape is continuous and not torn; re-tape gaps.
 3. Dim light? Battery might be weak; replace it.
 - **Group Work:** Encourage collaboration—students can help each other troubleshoot.

3. Exploration and Extension Activities (10-15 minutes)

- **Creative Application:** Once the basic circuit works, let students add art: Color around the LED to make it part of a scene (e.g., a firefly's glow).
- **Advanced Challenges:**
 - Add a Switch: Use a paper flap or binder clip to break/open the circuit.
 - Multiple LEDs: Create a parallel circuit (branching paths) for 2-3 lights.
 - Series Circuit: Connect LEDs end-to-end (note: may dim if voltage is insufficient).
 - Theme-Based Projects: Light-up holiday cards, solar system models (e.g., planets lighting up), or thaumatropes (spinning illusions with LEDs).
- **Integration Ideas:**
 - Art: Design thematic cards (e.g., birthdays).
 - Science: Test other conductors (foil vs. pencil graphite).
 - Math: Measure tape lengths or calculate simple resistance (for older students).

4. Wrap-Up and Assessment (5-10 minutes)

- **Reflection:** Have students share their creations in a gallery walk or quick presentations. Ask: "What worked well? What was challenging? How does this relate to real-world technology?"
- **Exit Ticket:** Students write or draw one thing they learned about circuits and one question they still have.
- **Formative Assessment:**
 - Observation: Check if circuits function and if students can explain components.
 - Rubric: Score on completion (does it light?), creativity, and troubleshooting (1-4 scale).
 - Optional Quiz: Simple questions like "What does the battery provide?" or "Why must the circuit be closed?"

Differentiation

- **For Younger Students (Grades 3-5):** Use pre-printed templates with guided lines; focus on simple on/off circuits.
- **For Older Students (Grades 6-8):** Introduce concepts like resistance, voltage drops, or parallel vs. series; challenge them to add sensors (e.g., paper pressure switches).
- **For Diverse Learners:** Provide visual aids, pair students, or offer pre-cut materials for those with fine motor challenges. Extend for gifted students with complex designs.

Resources and Extensions

- **Online Resources:** Chibitronics Educator Guide for advanced ideas; YouTube tutorials on simple paper circuits (e.g., light-up cards).
- **Follow-Up Lessons:** Explore squishy circuits (with playdough), wearable tech, or coding with micro:bit for integrated circuits.
- **Cost Estimate:** Low—under \$5 per student for reusable materials.
- **Potential Challenges:** Sticky tape can be tricky; have extras. Ensure inclusivity by grouping diverse skill levels.

This lesson fosters curiosity in electronics while blending creativity, making it an engaging entry into STEAM education. Students often leave excited to experiment at home!